YMnotes

CHAPTER 09

Parasitic GI Diseases

Blood flukes · tapeworms · intestinal nematodes · extra-intestinal amoebiasis — redesigned edition of the YMnotes chapter

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HOW TO READ THE COLOURS

- **Main text** · the YMnotes chapter content. Each parasite has its own accent colour.
- **Amber** · **Clinical clarification**: a source statement checked against current CDC / WHO guidance.
- **Brown note** · editorial note: an unclear or incomplete source line, kept and flagged.
- **Red** · high-yield points and complications.

IN THIS CHAPTER

- **4 parasite groups**, each with distribution, hosts, life cycle, pathogenesis, diagnosis, treatment and control.
- **7 figures**: life cycles of *Schistosoma*, *Taenia*, *Ascaris*, *Enterobius* and *E. histolytica*; *Ascaris* worms and egg.
- **Staged schistosomiasis** — invasion, migration, egg deposition, tissue reaction.
- A **quick comparison table** at the end for revision.

PART 1 · BLOOD FLUKES

1 *Schistosoma*

Schistosomes are the causative agents of the disease **schistosomiasis**. Three species of schistosomes that parasitize humans are of major importance: ***S. mansoni***, ***S. haematobium*** and ***S. japonicum***.

THE TWO INTESTINAL SPECIES COMPARED

	S. mansoni	*S. japonicum*
Disease	Intestinal bilharziasis	Intestinal bilharziasis
Distribution	Nile Delta, Africa, South America, Middle East	Far East
Intermediate host	*Biomphalaria alexandrina*	*Oncomelania* species
Reservoir host	Monkeys, rodents	Dogs, cats, rodents, pigs and cattle
Habitat	**Inferior** mesenteric plexus of veins	**Superior** mesenteric plexus of veins

EDITORIAL NOTE

The source introduces the table as covering "the three species", but it gives only *S. mansoni* and *S. japonicum* — the two intestinal species. No *S. haematobium* column is supplied, so none is added here.

Life cycle

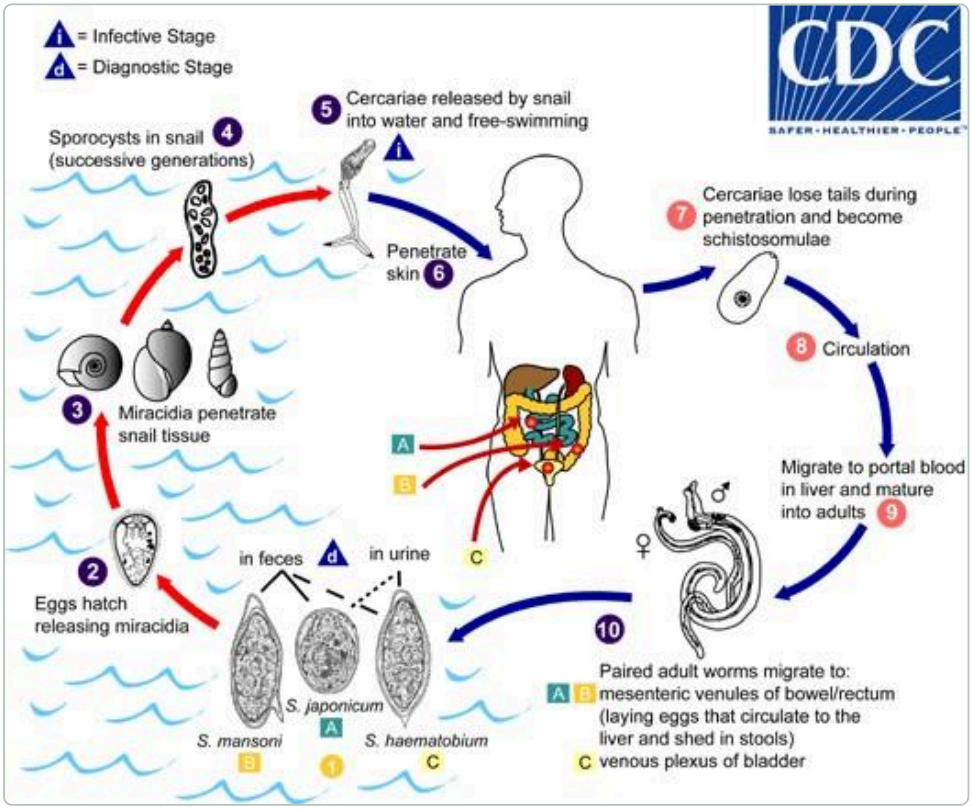


Figure 9.1 — *Schistosoma* life cycle · CDC

Eggs escape to the perivascular tissue due to

1. Pressure within the venule.
2. Effect of the **spine**.
3. The oozing **lytic secretion of the miracidium** within the egg.

From the perivascular tissue, eggs pass to the lumen of the **intestine** (*S. mansoni*, *S. japonicum*) or of the **urinary bladder** (*S. haematobium*).

Cercariae penetrate the skin, aided by

1. **Proteolytic enzymes** secreted from penetration glands.
2. The **surface tension** of drying droplets of water.
3. The strong **lashing movements of the tail** pressing the body of the cercaria into the skin.

When the **embryonated eggs (diagnostic stage)** discharged in faeces enter fresh water, **miracidia hatch** from the eggs and rapidly swim by means of cilia (they survive for **24 hours**). They then penetrate the body of the **snail (intermediate host)**.

Drinking water may lead to infection when cercariae penetrate the mucosa above the level of gastric acidity, which kills them. Cercariae lose their tails and transform into **schistosomula**.

MIGRATION INSIDE THE DEFINITIVE HOST — IN THE SOURCE'S ORDER

1. **Venous circulation** schistosomula migrate to the **lungs**
2. **Through the lungs** break out through the pulmonary capillaries; carried through the left heart into the systemic circulation
3. **Liver** reach the intra-hepatic branches of the **portal veins**, settle down and mature
4. **Pairing → egg laying** within several weeks, paired worms migrate to the **mesenteric veins** (*S. mansoni*, *S. japonicum*) or **vesical plexus** (*S. haematobium*)

Prepatent period (cercarial penetration → eggs in the excreta): about **35 days** for *S. mansoni* and *S. japonicum*, and about **70 days** for *S. haematobium*.

Pathogenesis and clinical picture

Intestinal schistosomiasis: **7–8 weeks** for *S. mansoni* infection and **5–6 weeks** for *S. japonicum* infection.

> EDITORIAL NOTE
>
> The source sentence reads "Intestinal schistosomiasis is 7-8 weeks ... and 5-6 weeks" without naming which period is meant. It is kept as printed rather than guessed.

Pathological changes are caused mainly by **schistosome eggs deposited in various tissues**. The antigens excreted from embryonated eggs result in formation of **granulomas** that lead eventually to **fibrotic changes** in the tissues involved in chronic infections. Thus, the lesions produced are essentially the same among the 3 major species. There are **4 progressive stages**:

1 · Invasion 1–4 days	Skin lesion due to cercarial penetration: **local dermatitis, irritation, papular rash**.

2 · Migration
3–4 weeks

- **Lung:** verminous pneumonitis, minute haemorrhages, cough and haemoptysis.
- **Liver:** enlarged and tender.

Metabolic products of migrating parasites → **toxic and allergic manifestations**: urticaria, **eosinophilia**, leukocytosis, fever, headache and muscle pain.

3 · Egg deposition & extrusion
acute stage · 1–2 months

Eggs deposited in the venous plexus escape from the veins into the perivascular tissue and finally find their way to the outside with urine and stool (**open-lesion stage**). With extrusion of eggs through the wall of the intestine or bladder there may again be generalized malaise, fever, rigors, urticaria, abdominal pain and liver tenderness.

Intestinal bilharziasis (*S. mansoni*, *S. japonicum*): active egg deposition → tissue damage and haemorrhage. Symptoms: **dysentery with blood and mucus in stool**, abdominal pain. *S. japonicum* may present with **Katayama fever**.

4 · Tissue reaction
chronic stage

Eggs trapped in the tissue secrete **soluble egg antigen** that stimulates inflammatory reactions (**eosinophilic granulomas**) and **fibrosis** → bilharzial nodules, papillomata, sandy patches and reduced egg output (**closed-lesion stage**).

Intestine (*S. mansoni*, *S. japonicum*): the wall becomes fibrosed and thickened and may be complicated with **strictures, sinuses, fistulae and prolapse**.

HEPATOSPLENIC DISEASE — THE SERIOUS END OF THE CHRONIC STAGE

Eggs that fail to be fixed in the venules of the intestinal wall fall into the lumen and are swept by the blood to the **liver** → **periportal fibrosis**, portal hypertension, hepatosplenomegaly, followed by a **shrunken fibrosed liver**.

Portal hypertension → oesophageal varices, haemorrhoids and ascites, manifested by **haematemesis, melena** and a distended abdomen. This stage ends in **hepatocellular failure** (impaired liver function) and fatal haemorrhage. The enlarged spleen and hypersplenism destroy red blood cells, adding to the existing anaemia; later this leads to **pancytopenia**.

Diagnosis of intestinal schistosomiasis

Approach	Details
1 · Clinical	Living in or visiting endemic areas with a history of exposure to **canal water**, and complaining of **dysentery or hepatosplenomegaly**, is suggestive.
2 · Parasitological	Stool examination for recovery of the eggs (diagnostic stage) by (a) **direct smear**; (b) **concentration techniques**, which may be required when infection is light.
3 · Immunodiagnosis	Detection of **antigens and antibodies**, as in urinary schistosomiasis.
4 · Other tests	(a) **CBC**: eosinophilia up to **40%** and anaemia. (b) Rectal swab. (c) **Colonoscopy and sigmoidoscopy** for visualizing lesions (e.g. papillomata) and for biopsies. (d) **Ultrasonography** and other imaging: periportal fibrosis and hepatosplenomegaly.

EDITORIAL NOTE

"As in urinary schistosomiasis" refers to material not included in this chapter. In item (c) the source also lists a lesion printed illegibly ("cologic peiyrosis"); it is omitted rather than guessed.

Treatment (as given in the source)

Praziquantel (Biltricide), **oxamniquine** (Vansil) and **metrifonate**.

Prevention and control

1. Mass treatment and follow-up of infected persons.
2. Health education.
3. Snail control.

CLINICAL CLARIFICATION · TREATMENT

Praziquantel remains the drug of choice for schistosomiasis and is effective against all major *Schistosoma* species (CDC). Oxamniquine acts only on *S. mansoni* and is no longer readily available; metrifonate is active only against *S. haematobium*, so it does not treat intestinal disease.

Sources: CDC, Clinical Care of Schistosomiasis; Colley DG, "Human schistosomiasis" (CDC Stacks).

PART 2 · TAPEWORMS

2 *Taenia saginata* — beef tapeworm

Also called the **bald tapeworm**. Cosmopolitan, especially in **cattle-raising countries**.

Habitat	Small intestine
Definitive host	**Man** (the only definitive host)
Intermediate host	Cattle
Infective stage	**Cysticercus bovis** in cattle meat
Diagnostic stage	Mature egg in stool

Mode of infection

Ingestion of **undercooked beef** containing viable cysticerci, usually as hamburgers or grilled meat (kebab). In the small intestine the **scolex is evaginated**, attaches to the mucosa and **matures in about 3 months**. The adult lives in the small intestine, attached to the mucosa.

Pathogenesis and clinical picture

1. Intestinal disturbances, **hunger pains** and diarrhoea.
2. Neurologic manifestations.
3. Intestinal obstruction.
4. **Loss of weight** — the parasite consumes much of the patient's food, leading to malnutrition and avitaminosis.
5. **Appendicitis or cholangitis** caused by stray segments of the worm.
6. **Migrating segments creeping out of the anus** cause irritation, itching and worry.

Diagnosis

1 · Clinical	Suggestive symptoms such as diarrhoea and weight loss, with **passage of gravid segments through the anus**.
2 · Parasitological	Stool examination for **eggs and proglottids** (diagnostic stages). Recovery of eggs from the perianal region using a **swab**. Gravid segments in faeces — if not found, give a **saline purge**. Compress the segments between two slides and **count the main lateral branches**.
3 · Immunodiagnosis	Detection of **coproantigen** in stool by ELISA.

Treatment

1. **Praziquantel** is the drug of choice for both *T. saginata* and *T. solium*; it causes worm paralysis — **10 mg/kg as a single oral dose**.
2. **Yomesan** (niclosamide): a single oral dose of **2 g** on an empty stomach.

Control

1. Treatment of infected persons.
2. Avoid grazing cattle on fields fertilized by human manure.
3. Proper meat inspection in slaughterhouses.
4. Refrigeration of meat at **−10 °C for 5 days**.
5. Thorough cooking of beef.

CLINICAL CLARIFICATION · TAENIASIS TREATMENT

CDC gives praziquantel **5–10 mg/kg orally once** (10 mg/kg may cure more often) and niclosamide **2 g once** as an alternative; WHO lists the same single-dose options. CDC advises using praziquantel and albendazole **carefully in anyone suspected of having cysticercosis** (seizures have been reported) — relevant to *T. solium* below.

Sources: CDC, Clinical Care of Taeniasis; WHO, Taeniasis/cysticercosis fact sheet. The −10 °C / 5-day freezing figure was not independently re-verified.

2.2 *Taenia solium* — pork tapeworm

Cosmopolitan where pork is eaten; thus it is **rare in Islamic and Jewish communities**. Similar to *T. saginata*, but differs in:

Feature	*T. solium*	*T. saginata* (for contrast, from above)
Intermediate host	**Pig**	Cattle
Larval stage (infective)	**Cysticercus cellulosae** — invaginated scolex with **four suckers and hooks**	Cysticercus bovis
Infection by	Ingestion of **undercooked pork**	Undercooked beef

Pathogenesis and clinical picture

1. Intestinal disturbances, neurologic manifestations, intestinal obstruction or loss of weight.
2. When man **ingests eggs** of *T. solium*, the larval stage (cysticercus cellulosae) develops in extra-intestinal tissues → **cysticercosis**. This occurs with *T. solium*, not with *T. saginata* — which makes *T. solium* run a more serious course.

Diagnosis

1. Clinical diagnosis.
2. Laboratory: stool examination for the **gravid segments**. Eggs are rarely present in stool and are similar to those of *T. saginata*.

Treatment (as given in the source)

1. Similar to that of *T. saginata*.
2. **Yomesan** 2 g single oral dose on an empty stomach, then a **saline purge 1–2 hours later** to avoid cysticercosis from eggs of disintegrated segments. Quinacrine hydrochloride (atebrine) can be added, as it expels the parasite intact.
3. Give an **antiemetic** before treatment to avoid vomiting.

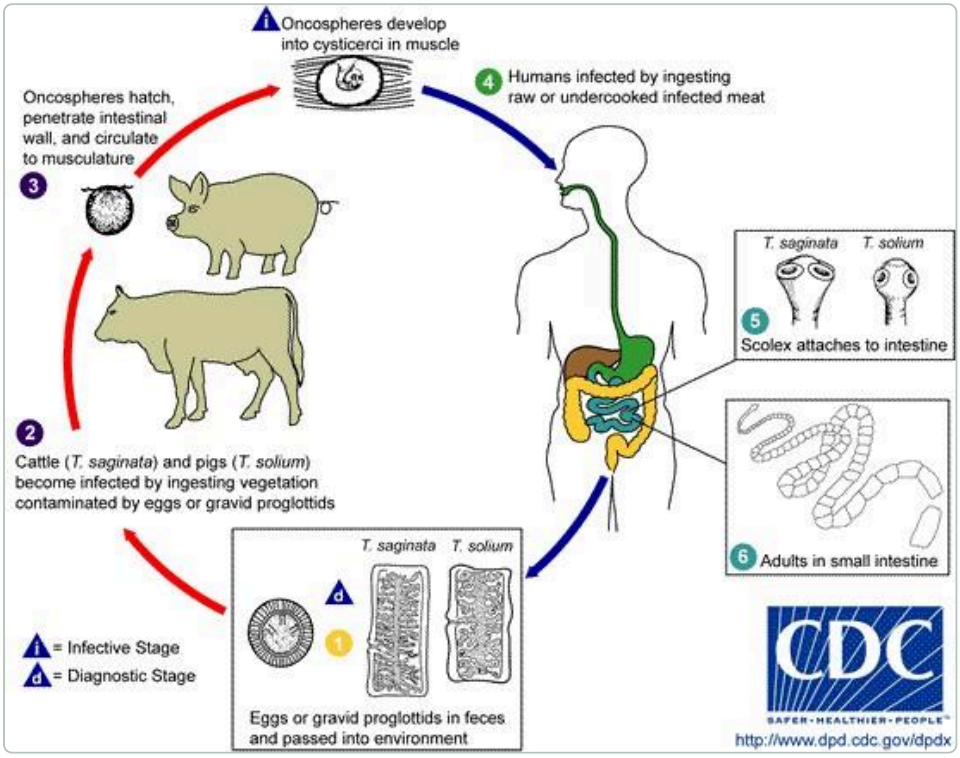


Figure 9.2 — *Taenia saginata* and *T. solium* life cycle · CDC

PART 3 · INTESTINAL NEMATODES

3 *Ascaris lumbricoides* — giant intestinal roundworm

Distribution	Cosmopolitan; more prevalent in warm countries with bad sanitation. **Common in Egypt** (especially children).
Host	Man
Habitat	Small intestine (**free in the lumen**)
Life cycle	**Geohelminthic** infection

Infection

Infection of man takes place by swallowing water or raw vegetables polluted with **embryonated eggs containing the infective larva**, or through hands contaminated by polluted soil.

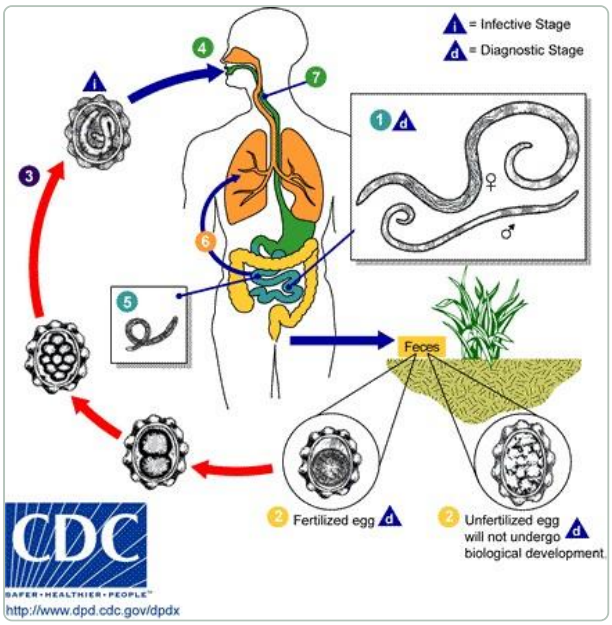


Figure 9.3 — *Ascaris* life cycle · CDC

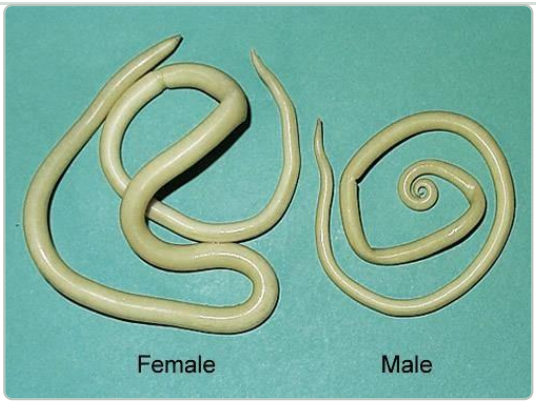


Figure 9.4 — *Ascaris*: male and female worms



Figure 9.5 — *Ascaris* egg

Pathogenesis and clinical picture

Tissue damage may be due to: (1) the **large size** of *Ascaris* (largest intestinal nematode); (2) adults **do not attach** to the intestinal wall, so they may go to ectopic places; (3) toxic products stimulate an immune response.

The usual infection (**10–20 worms**) often passes unnoticed. In heavier infection:

1 · During larval migration

Intense local inflammatory reaction around the larvae → **pneumonitis**: fever, cough, dyspnoea and **eosinophilia**.

In the intestine

Abdominal pain, nausea, vomiting, colic, distension or dyspepsia, due to production of **anti-enzymes** that interfere with protein digestion → malnutrition and **underdevelopment of children**. Changes of bowel habit (diarrhoea or constipation).

2 · COMPLICATIONS

Traumatic effects — adult worms irritated by **fever, drugs or anaesthesia**. A mass of worms causes **intestinal obstruction, volvulus and intussusception**. Worms are not fixed, so they may wander to:

Bile ducts → obstructive jaundice	**Liver** (carrying intestinal bacteria) → liver abscesses
Appendix → appendicitis	**Ampulla of Vater** → acute haemorrhagic pancreatitis
Peritoneum → peritonitis	**Stomach** → vomiting, or escape through the external nares
Trachea (very rarely) → suffocation	

Toxic effects — by-products of living and dead worms in sensitized persons: oedema of the face, urticaria, asthma, insomnia, nervous irritability and convulsions.

Diagnosis, treatment and prevention

Diagnosis

1 · Clinical: non-specific.
2 · Parasitological: **eggs in stool**; **larvae in sputum** and **Charcot–Leyden crystals** early in the disease.
3 · Other: **high eosinophilia**; X-ray after barium meal shows worms as **filling defects**.

Treatment

Mebendazole or flubendazole: **100 mg twice daily for 3 days**.

Prevention and control

1 · Purification and chlorination of water. 2 · Washing raw vegetables; washing hands before meals. 3 · Proper disposal of sewage (human faeces). 4 · Mass treatment, personal hygiene and education.

3.2 *Enterobius vermicularis* — pinworm, seatworm

Distribution	Cosmopolitan; more in **temperate and cold** climates
Final host	Man, especially **children**
Habitat	**Large intestine**: mainly caecum, colon, rectum and appendix

Mode of infection

1. Ingestion of infective eggs through **autoinfection** (contaminated hand to mouth) or contaminated food or drinks.
2. **Airborne** infection through inhalation of eggs in dust particles.

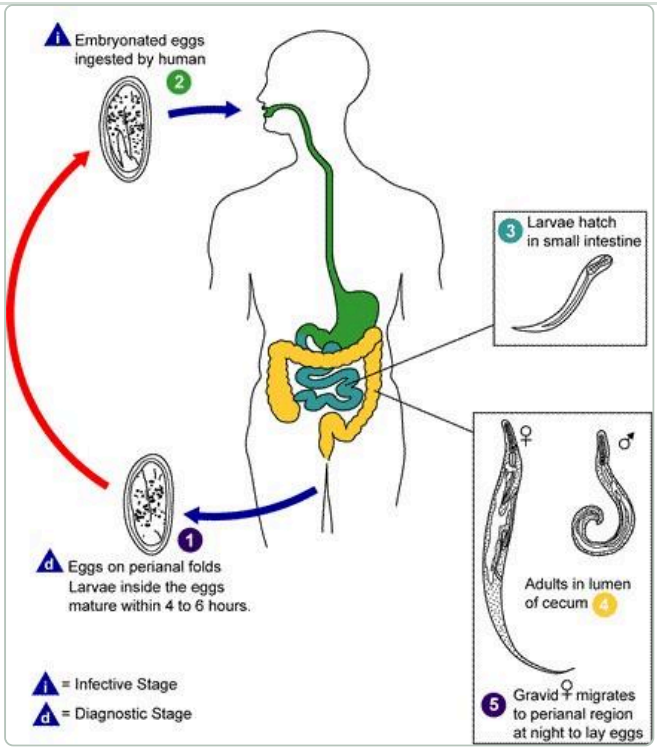


Figure 9.6 — *Enterobius vermicularis* life cycle · CDC

Pathogenesis and symptoms

1. **Irritation, itching and scratching of the perianal skin (pruritus ani)** → insomnia, nervous manifestations and secondary bacterial infection.
2. Worms in young girls may deposit eggs on the vulva → **pruritus vulvae**.
3. Worms may migrate to the vagina (vaginitis), uterus and urinary bladder (urethritis), causing irritation and inflammation.
4. **Appendicitis** due to migrating adults.
5. Because infection spreads easily within a family, **all the family should be examined and all infected members treated at the same time**.
6. Sometimes the vagina and bladder are invaded in female patients, causing irritation.
7. Migration to the peritoneal cavity (through the uterine tubes) causes **pelvic peritonitis**.

Diagnosis

Clinically: suspected in children with **pruritus ani at night**.
Laboratory: adult worms may be seen in stool or the anal area. **Eggs are rarely found in stool (about 5% only)**. Swab the anal / perianal area with (a) an **NIH swab** (National Institutes of Health) or (b) a **Scotch adhesive-tape swab**.

Treatment (as given in the source)

1 · **Albendazole, mebendazole**.
2 · Local application of white oxide of mercury (white precipitate ointment) around the anus at night (relieves itching, kills female worms coming out, prevents dispersal of eggs).

Prevention and control

Personal hygiene. Infected persons should sleep alone; their underwear and bedding handled and washed carefully. Food and drink protected from dust and hands.

CLINICAL CLARIFICATION · PINWORM TREATMENT

CDC's regimen is a **single dose repeated after 2 weeks** — mebendazole 100 mg, albendazole 400 mg, or pyrantel pamoate 11 mg/kg (max 1 g) — because the drugs kill worms but **not eggs**. Household members are treated together when more than one is infected or infection recurs. The mercury ointment in item 2 is **not part of current CDC treatment guidance**.

Sources: CDC, Clinical Overview of Pinworm Infection; CDC Yellow Book 2024, Enterobiasis.

PART 4 · EXTRA-INTESTINAL AMOEBIASIS

4 Hepatic amoebiasis

Pathogenesis

a · Direct spread

Of amoebic **trophozoites** from a primary intestinal lesion to surrounding tissues.

b · Haematogenous spread

Through invasion of blood: the **portal circulation** draining the intestine carries amoebic emboli mainly to the **right lobe of the liver**.

Liver affection may take the form of **diffuse amoebic hepatitis** or **liver abscess**.

Diffuse amoebic hepatitis	In a considerable number of patients with amoebic colitis there is an associated **enlargement of the liver**. It starts with constant invasion of the liver by amoebae from an active intestinal process via the portal circulation.
Amoebic liver abscess	Where amoebic colonization continues there is **lytic necrosis** of hepatic parenchyma. Lesions coalesce, typically forming a **solitary** lesion that begins as a small area of necrosis and enlarges, with liquefaction in the centre and thickening of the wall. Contents: lysed liver cells, bile, fat, RBCs and collagen strands — a viscid chocolate-red or creamy mass (**"anchovy-sauce" pus**). **Trophozoites are found only in the edge** of the lesion, where they multiply. Commonest site: the **posterior part of the right lobe (silent area)**, causing upward bulging of the diaphragm. The abscess may become chronic, absorbed, fibrosed, calcified, or extend to other organs.

Clinical picture

Symptoms

Usually an **acute onset of upper-abdominal pain with high fever**. The pain is intense and constant, **radiating to the scapular region and right shoulder**, increasing with deep breathing or coughing. Fever **38–40 °C**; anorexia, nausea, vomiting and fatigue.

Signs

1. **Enlarged tender liver**
2. Pallor
3. Earthy look
4. Fever
5. Tinge of jaundice

Differential diagnosis: other causes of hepatomegaly and cystic lesions in the liver (**hydatid disease**).

Diagnosis

a · Clinical	The clinical picture.
b · Laboratory	**Indirect (serology)** for specific antibodies: **IHAT** (indirect haemagglutination) and **ELISA** are sensitive and specific but **cannot assess cure**, because antibodies persist for a long time. **Direct:** (i) **aspiration** of the abscess — specimen from the **border** for trophozoites; (ii) stool examination — positive in only **one third** of cases; (iii) **copro-antigen** detection by enzyme immunoassay.
c · Radiology	X-ray: high density in the hepatic shadow and **abnormal elevation of the right dome of the diaphragm**. Ultrasonography, computed tomography (CT) and magnetic resonance may visualize the abscess.
d · CBC	**Polymorphonuclear leucocytosis** in amoebic liver abscess (differentiates it from viral hepatitis).

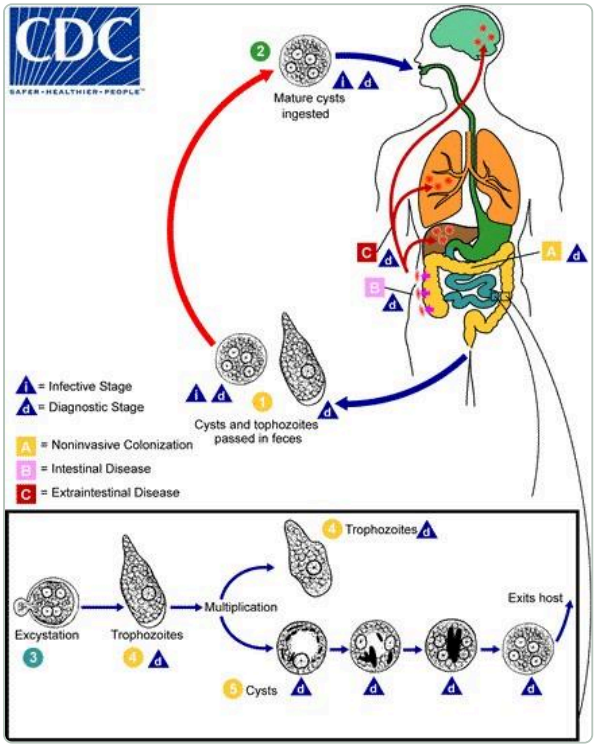


Figure 9.7 — *Entamoeba histolytica* life cycle · CDC

Prevention and control

1. Personal hygiene.
2. Washing fresh-eaten vegetables and fruits.
3. Safe water supply; water from a polluted source should be boiled.
4. Insect control, or protection of food from insects.
5. Treatment of patients and examination of contacts.

The source chapter gives no treatment for hepatic amoebiasis; none is added here.

STUDY AID

Quick comparison

Compiled only from this chapter's statements. A dash means the chapter gives no entry.

Parasite	Habitat in man	How man is infected	Diagnostic stage / key test	Treatment named
S. mansoni* / *S. japonicum	Inferior / superior mesenteric veins	Cercariae penetrate skin (or mucosa in drinking water)	Embryonated egg in stool; CBC eosinophilia up to 40%	Praziquantel (drug of choice)
T. saginata	Small intestine	Undercooked beef (cysticercus bovis)	Egg / gravid segments; count lateral branches; coproantigen	Praziquantel 10 mg/kg once; niclosamide 2 g
T. solium	Small intestine	Undercooked pork (cysticercus cellulosae); eggs → cysticercosis	Gravid segments (eggs rare)	As *T. saginata* + saline purge; antiemetic
Ascaris	Small intestine, free in lumen	Embryonated eggs in water, raw vegetables, soil on hands	Eggs in stool; larvae + Charcot–Leyden crystals in sputum early	Mebendazole / flubendazole 100 mg bid × 3 days
Enterobius	Caecum, colon, rectum, appendix	Autoinfection, food/drink, inhaled eggs	NIH / Scotch-tape swab (eggs in stool only ~5%)	Albendazole, mebendazole (repeat after 2 weeks — CDC)
***E. histolytica* · liver**	Liver, mainly right lobe (posterior)	Spread from intestinal lesion via portal circulation	Serology (IHAT, ELISA); aspirate from abscess border	—

PRESENTATION CHANGES ONLY

This is a redesign of the YMnotes chapter (printed pages 122–139). Obvious typing and OCR slips were corrected without changing meaning — e.g. "Blood flukes", "superior mesenteric", "eosinophilic", "fistulae", "ascites", "praziquantel", "oxamniquine", "metrifonate", "*lumbricoides*", "pruritus", and fever "38–40 °C" (printed "380 and 400 C"). Unclear lines are kept and flagged, not resolved from memory. Only the amber clinical clarifications were checked against CDC and WHO; figures are reproduced from the source at their original resolution.

رَبِّ زِدْنِي عِلْمًا

[سورة طه : 114]

ymnotes Medical

محاضرات فيديو

مراجعة شاملة

COMPLETE MEDICAL VIDEO LECTURES

- ENT
- Forensic Medicine
- Special Sense Eye
- Ophthalmology
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- Cardiovascular System
- Cardiology
- Respiratory System
- Chest
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- General Surgery
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شرح مبسط ومنظم | محتوى شامل ومحدث | مناسب لجميع مراحل الدراسة | راجع في أي وقت ومن أي مكان